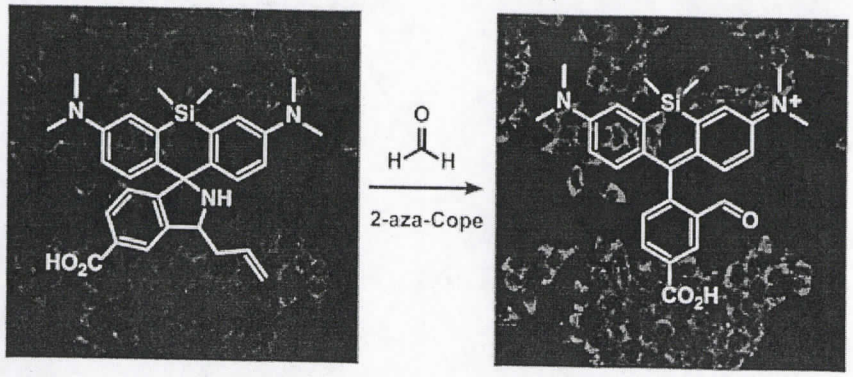
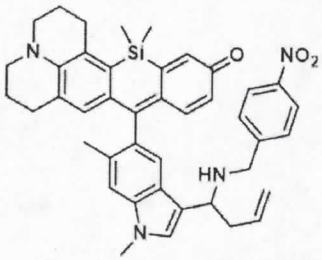
Organic Cumulative Exam (topic: 2- Aza-Cope rearrangement)
March 11, 2016

1. (70 points): Propose a reasonable mechanism for the following transformation. (*J. Am. Chem. Soc.* **2015**, *137*, 10886–10889).



2. (30 points) Similar to the above example, the following compound can also be used to detect cellular formaldehyde through 2-aza-cope rearrangement. Suggest all the products, **in particular by-product(s)**, after FP1 reacts with formaldehyde. (*J. Am. Chem. Soc.* **2015**, *137*, 10890–10893).



FP1